

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

7 Hypothesis Testing

Unlike situations where we directly estimate some population parameters, sometimes we may have specific claims about some unknown population parameters - and, instead of the exact true values of parameters, we may care more about whether those claims of interest are true or false. In this case, we use some sample data to investigate the claims.

7.1 Basics

Step 1: Population

The very first step of a hypothesis test is specifying a probability model for the sample observations - i.e., making assumptions on the distribution followed by the population.

- In the context of this course, we normally either consider a normal population or work with large sample.

Step 2: Hypothesis

A hypothesis test is an investigation on values of population parameters. Thus, we need to figure out the set of possible values of the parameters so that we can then define hypotheses accordingly.

Definition 7.1.1. For a population parameter, the **parameter space** is the set of possible values of the parameter.

Definition 7.1.2.0. A **statistical hypothesis** (or just hypothesis) is a claim or assertion about one of the following things:

- the value of a single population parameter;
- the values of several population parameters;
- the form of an entire probability distribution (we won't discuss this type of hypotheses in this course).

Definition 7.1.2.1. A **null hypothesis**, denoted by H_0 , is the claim that is initially assumed to be true - in some sense, the “prior belief” claim.

- H_0 will be rejected only if sample evidence strongly contradicts H_0 and suggests that H_0 is false. Otherwise, we will continue to believe in the plausibility of H_0 .
- Accordingly, the two possible conclusions drawn in a hypothesis test are “reject H_0 (the null)” and “fail to reject H_0 (the null)”.

Definition 7.1.2.2. An **alternative hypothesis**, denoted by H_A , is the claim that is contradictory to H_0 .

- A pair of null and alternative hypotheses about parameter values often - though not always - partition the parameter space into two subsets, which, in this case, are respectively claimed by the two hypotheses.

Take the case with a single parameter $\mu \in \mathbb{R}$ as an example:

- $H_0 : \mu = \mu_0$ & $H_A : \mu \neq \mu_0$ and $H_0 : \mu \geq \mu_0$ & $H_A : \mu < \mu_0$ are cases where the parameter space is partitioned by the two hypotheses;
- $H_0 : \mu = \mu_0$ & $H_A : \mu < \mu_0$ is a case where the parameter space is not partitioned by the two hypotheses.

Definition 7.1.3.1. A hypothesis is called a **simple hypothesis** if it claims single values for parameters.

Definition 7.1.3.2. A hypothesis is called a **composite hypothesis** if it claims that the true values of parameters fall in a larger set (usually intervals in the context of real numbers).

Step 3: Test Statistic

With hypotheses ready, our next step is to use sample data in calculation to conduct the test.

Definition 7.1.4. A **test statistic** is a function of the sample data used as a basis for deciding whether H_0 should be rejected.

- Because the test statistic is a function of the sample, thus being a r.v., its realization (value calculated from the available sample data) is with some randomness. Therefore, we set some conditions so that we reject the null if and only if the conditions are satisfied by the realization of the statistic.
- Such conditions partition the set of possible realizations of the test statistic into two parts, namely **acceptance region** and **rejection region** - so that
 - if the realization falls into the rejection region, we would reject the null, and
 - if, instead, the realization falls into the acceptance region, we would fail to reject the null.

- Due to the randomness of the test statistic's realization, there are two possible types of errors that may happen when drawing a conclusion in a hypothesis test: rejecting the null when it is actually true, and failing to reject the null when it is actually false.

Definition 7.1.5.1. A **type I error**, whose probability is conventionally denoted as α , occurs in a hypothesis test if a true H_0 is rejected.

- In a given test, there is a value of α for each value of the parameter in H_0 . Thus, there is a value of parameter in H_0 with the largest α .

Definition 7.1.5.2. The maximum probability of type I error in a test is called the **significance level** (or **size**) of the test.

- Normally we only care about the significance level of the test rather than the probability of type I error for each value covered in H_0 of a test. Thus, the (value of) significance level of a test is also usually denoted as α .
- The significance level here should be considered as statistical significance, which should be distinguished from practical significance.
 - Statistical significance evaluates how strong the conclusion (of, in our context here, a hypothesis test) is - or in other words, how confident a statistician is in the validity of the conclusion.
 - By contrast, practical significance is about how meaningful the conclusion (of the test) is in the real world.
 - * For example, with a large sample we may - from a sample mean whose value is 151 (lbs) - reject that the average weight of the population is 150 lbs. However, the 1 lb difference between the sample mean and the null may not be such a big deal in practice, and in such situations it may not be preferable for us to have the sample evidence argue strongly for the rejection of H_0 .

Definition 7.1.5.3. A **type II error**, whose probability is conventionally denoted as β , occurs in a hypothesis test if a false H_0 is not rejected.

- By contrast, $1 - \beta$, i.e., the probability of correctly rejecting H_0 (i.e., rejecting H_0 when it is false) in a test, is called the **power** of the test.

Proposition 7.1.6. In a given hypothesis test, with everything else (e.g., sampling procedure, sample size, test statistic) held fixed, there exists some pair of acceptance region and rejection region with a pair of (α, β) that can't be both further reduced by any other pair of acceptance region and rejection region.

- Proof: not required.
- Such an “optimal” pair of acceptance region and rejection region is usually - at least when the sampling distribution of the test statistic is symmetric and unimodal (e.g., normal or t distribution) - with the acceptance region being a single interval.

Following the discussion above, there are two ways of proceeding with the hypothesis test.

Step 4 Version 1: Significance Level - Rejection Region

Normally, we pick a significance level α that we can accept, and then derive the rejection and acceptance regions accordingly.

- The rejection region is such that

$$P(\text{The realization of the test statistic falls in the rejection region} | H_0 \text{ is true}) = \alpha.$$

Step 4 Version 2: Rejection Region - Significance Level

Alternatively, we may continue by first specifying the rejection and acceptance regions and then calculate α according to the sampling distribution of the test statistic.

Besides, there are two other methods of hypothesis testing that more or less depart from the standard procedure described above.

P-value Method

Definition 7.1.7. A **p-value** is the probability, calculated assuming that H_0 is true, of obtaining a value of the test statistic at least as contradictory to H_0 as (or equivalently, more extreme than) the value calculated from the available sample data.

- Mathematically, in the context of real numbers,

$$\text{p-value} = \begin{cases} P(\hat{\theta} > \tilde{\theta}) & \text{if } H_0 \text{ is one-sided and states that } \theta \text{ is small, e.g., } \theta \leq \theta_0 \\ P(\hat{\theta} < \tilde{\theta}) & \text{if } H_0 \text{ is one-sided and states that } \theta \text{ is large, e.g., } \theta \geq \theta_0 \\ P(|\hat{\theta} - \theta_0| > |\tilde{\theta} - \theta_0|) & \text{if } H_0 \text{ is "two-sided", e.g., } \theta = \theta_0 \end{cases},$$

where θ is the parameter associated with the test, θ_0 is the parameter value involved in H_0 , $\hat{\theta}$ is the estimator - which is therefore a r.v. - of the parameter, $\tilde{\theta}$ is the value of the parameter's estimate calculated from the available sample data.

- The p-value is the maximum significance level at which one may fail to reject the null. The smaller the p-value, the stronger is the evidence from the sample data against H_0 .
- When H_0 is a composite hypothesis, the p-value should then be the maximum of the probabilities of estimator's being more extreme than the available estimate with each parameter value covered in H_0 assumed to be true, e.g.,

$$\text{p-value} = \begin{cases} \max_{\theta' \leq \theta_0} P(\hat{\theta} > \tilde{\theta} | \theta = \theta') & \text{if } H_0 : \theta \leq \theta_0 \\ \max_{\theta' \geq \theta_0} P(\hat{\theta} < \tilde{\theta} | \theta = \theta') & \text{if } H_0 : \theta \geq \theta_0 \end{cases}.$$

Confidence Interval Method

To do a test with significance level $1 - \alpha$, instead of constructing rejection and acceptance regions regarding the (critical) value of the parameter covered in H_0 , we may construct a $1 - \alpha$ confidence interval (either one-sided or both two-sided and symmetric, depending on the form of H_0) for the parameter.

- Then, we reject the null if and only if the confidence interval doesn't contain the above (critical) value.

7.2 One-Sample Hypothesis Testing

In the following sections, we discuss methods of doing different kinds of hypothesis tests at significance level α with random samples (which is, in one-sample situations, of size n , and, in two-sample situations, of sizes n_1 and n_2).

Method 7.2.1. Test about mean of a normal population $N(\mu, \sigma^2)$ with σ^2 known.

1. Here we work on the following three pairs of hypotheses:

- (a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0,$
- (b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0,$ and
- (c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0.$

2. By property of random sample and linear combination of normal r.v.'s,

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1).$$

3. For each pair of hypotheses, the rejection region - which is the complement of its corresponding acceptance region - is constructed in the following manner (with cv 's standing for critical values):

- (a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0.$

$$\begin{aligned} \frac{\alpha}{2} &= P(\bar{X} < cv_1 | \mu = \mu_0) = P(\bar{X} > cv_2 | \mu = \mu_0) \\ &= P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{cv_1 - \mu}{\frac{\sigma}{\sqrt{n}}} | \mu = \mu_0\right) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{cv_2 - \mu}{\frac{\sigma}{\sqrt{n}}} | \mu = \mu_0\right), \text{ yielding} \\ \frac{cv_1 - \mu_0}{\frac{\sigma}{\sqrt{n}}} &= z_{1-\frac{\alpha}{2}} = -z_{\frac{\alpha}{2}}, \quad \frac{cv_2 - \mu_0}{\frac{\sigma}{\sqrt{n}}} = z_{\frac{\alpha}{2}}, \text{ and thus} \\ cv_1 &= \mu_0 - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \quad cv_2 = \mu_0 + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \text{ constituting the} \\ \text{rejection region: } &(-\infty, \mu_0 - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}) \cup (\mu_0 + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, +\infty). \end{aligned}$$

- (b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0.$

$$\begin{aligned} \alpha &= \max_{\mu' \leq \mu_0} P(\bar{X} > cv | \mu = \mu') \text{ [often written as } P(\bar{X} > cv | \mu \leq \mu_0)] \\ &= \max_{\mu \leq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{cv - \mu}{\frac{\sigma}{\sqrt{n}}}\right), \text{ which, for each value of } \mu, \text{ corresponds to} \\ \frac{cv - \mu}{\frac{\sigma}{\sqrt{n}}} &= z_\alpha, \text{ constituting the} \\ \text{rejection region: } &\bigcap_{\mu \leq \mu_0} (\mu + z_\alpha \frac{\sigma}{\sqrt{n}}, +\infty) = (\mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}, +\infty). \end{aligned}$$

(c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0$.

$$\begin{aligned}
\alpha &= \max_{\mu' \geq \mu_0} P(\bar{X} < cv | \mu = \mu') \\
&= \max_{\mu \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{cv - \mu}{\frac{\sigma}{\sqrt{n}}}\right), \text{ which, for each value of } \mu, \text{ corresponds to} \\
&\frac{cv - \mu}{\frac{\sigma}{\sqrt{n}}} = z_{1-\alpha} = -z_\alpha, \text{ constituting the} \\
&\text{rejection region: } \bigcap_{\mu \geq \mu_0} (-\infty, \mu - z_\alpha \frac{\sigma}{\sqrt{n}}) = (-\infty, \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}}).
\end{aligned}$$

• Alternatively, we may head to the p-values (with $\bar{x}$ standing for sample mean):

(a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0$.

$$\begin{aligned}
\text{p-value} &= P(|\bar{X} - \mu| > |\bar{x} - \mu| | \mu = \mu_0) \\
&= P\left(\frac{|\bar{X} - \mu_0|}{\frac{\sigma}{\sqrt{n}}} > \frac{|\bar{x} - \mu_0|}{\frac{\sigma}{\sqrt{n}}}\right) \\
&= P\left(\frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} > \frac{|\bar{x} - \mu_0|}{\frac{\sigma}{\sqrt{n}}}\right) + P\left(\frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} < -\frac{|\bar{x} - \mu_0|}{\frac{\sigma}{\sqrt{n}}}\right), \text{ leading to} \\
z_{\frac{\text{p-value}}{2}} &= \frac{|\bar{x} - \mu_0|}{\frac{\sigma}{\sqrt{n}}}.
\end{aligned}$$

(b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0$.

$$\begin{aligned}
\text{p-value} &= \max_{\mu' \leq \mu_0} P(\bar{X} > \bar{x} | \mu = \mu') \\
&= \max_{\mu \leq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}\right), \text{ leading to} \\
z_{\text{p-value}} &= \min_{\mu \leq \mu_0} \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \text{ (as the critical value } z_q \text{ is decreasing in } q).
\end{aligned}$$

(c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0$.

$$\begin{aligned}
\text{p-value} &= \max_{\mu' \geq \mu_0} P(\bar{X} < \bar{x} | \mu = \mu') \\
&= \max_{\mu \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}\right) \\
&= \max_{\mu \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{\mu - \bar{x}}{\frac{\sigma}{\sqrt{n}}}\right), \text{ leading to} \\
z_{\text{p-value}} &= \min_{\mu \geq \mu_0} \frac{\mu - \bar{x}}{\frac{\sigma}{\sqrt{n}}} = \frac{\mu_0 - \bar{x}}{\frac{\sigma}{\sqrt{n}}}.
\end{aligned}$$

Method 7.2.2. Test about mean of a population with mean μ , unknown variance σ^2 , and large sample size.

1. Similar to the previous situation, we work on the following three pairs of hypotheses:

- (a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0,$
- (b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0,$ and
- (c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0.$

2. By a CLT, approximately (or, rigorously, in the limit),

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim N(0, 1).$$

3. For each pair of hypotheses, the rejection region is constructed in the following manner (with cv 's standing for critical values):

- (a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0.$

$$\begin{aligned} \frac{\alpha}{2} &= P(\bar{X} < cv_1 | \mu = \mu_0) = P(\bar{X} > cv_2 | \mu = \mu_0) \\ &= P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} < \frac{cv_1 - \mu}{\frac{S}{\sqrt{n}}} | \mu = \mu_0\right) = P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} > \frac{cv_2 - \mu}{\frac{S}{\sqrt{n}}} | \mu = \mu_0\right), \text{ constituting the} \\ &\text{rejection region: } (-\infty, \mu_0 - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}) \cup (\mu_0 + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}, +\infty). \end{aligned}$$

- (b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0.$

$$\begin{aligned} \alpha &= \max_{\mu' \leq \mu_0} P(\bar{X} > cv | \mu = \mu') \\ &= \max_{\mu \leq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} > \frac{cv - \mu}{\frac{S}{\sqrt{n}}}\right), \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\mu \leq \mu_0} (\mu + z_{\alpha} \frac{S}{\sqrt{n}}, +\infty) = (\mu_0 + z_{\alpha} \frac{S}{\sqrt{n}}, +\infty). \end{aligned}$$

- (c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0.$

$$\begin{aligned} \alpha &= \max_{\mu' \geq \mu_0} P(\bar{X} < cv | \mu = \mu') \\ &= \max_{\mu \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} < \frac{cv - \mu}{\frac{S}{\sqrt{n}}}\right), \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\mu \geq \mu_0} (-\infty, \mu - z_{\alpha} \frac{S}{\sqrt{n}}) = (-\infty, \mu_0 - z_{\alpha} \frac{S}{\sqrt{n}}). \end{aligned}$$

• Alternatively, p-values (with $\bar{x}$ as sample mean and s as sample standard error):

- (a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0.$

$$\begin{aligned} \text{p-value} &= P\left(\frac{|\bar{X} - \mu|}{\frac{S}{\sqrt{n}}} > \frac{|\bar{x} - \mu|}{\frac{s}{\sqrt{n}}} | \mu = \mu_0\right), \text{ leading to} \\ z_{\frac{\text{p-value}}{2}} &= \frac{|\bar{x} - \mu_0|}{\frac{s}{\sqrt{n}}}. \end{aligned}$$

(b) $H_0 : \mu \leq \mu_0, H_A : \mu > \mu_0$.

$$\begin{aligned} \text{p-value} &= \max_{\mu' \leq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} > \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \mid \mu = \mu'\right), \text{ leading to} \\ z_{\text{p-value}} &= \min_{\mu \leq \mu_0} \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}. \end{aligned}$$

(c) $H_0 : \mu \geq \mu_0, H_A : \mu < \mu_0$.

$$\begin{aligned} \text{p-value} &= \max_{\mu' \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} < \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \mid \mu = \mu'\right), \text{ leading to} \\ z_{\text{p-value}} &= \min_{\mu \geq \mu_0} \frac{\mu - \bar{x}}{\frac{s}{\sqrt{n}}} = \frac{\mu_0 - \bar{x}}{\frac{s}{\sqrt{n}}}. \end{aligned}$$

- With large sample, the population need not be normal for one to apply the method above.
 - For example, for a Bernoulli population with success probability p , one can start a test with

$$\frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0, 1).$$

- * Again, recall that a rule of thumb of applying such a CLT to a Bernoulli population is $np \geq 10$ and $n(1-p) \geq 10$.
- * One thing that differs in the CI construction and hypothesis test in the context of Bernoulli population is that the denominator of the statistic from the CLT consists of the estimator of mean (proportion) in the case of CI but only involves true parameter value in the case of hypothesis testing.
 - Though we still don't inherently know the true value of proportion in hypothesis testing, we assume some true value(s) in H_0 - and rejection regions as well as p-values are all calculated with H_0 assumed to be true.
 - * Another noteworthy fact in this situation is that the maximum probability of type I error (i.e., the significance level) may not be attained at the boundary between H_0 and H_A .
- If the population variance σ^2 is known in a large-sample situation, one can still apply the above method with each s directly replaced with σ .

Method 7.2.3. Test about mean of a normal population $N(\mu, \sigma^2)$ with σ^2 unknown and small sample size.

1. Again, we work on the following three pairs of hypotheses:

(a) $H_0 : \mu = \mu_0, H_A : \mu \neq \mu_0$,

- (b) $H_0 : \mu \leq \mu_0$, $H_A : \mu > \mu_0$, and
(c) $H_0 : \mu \geq \mu_0$, $H_A : \mu < \mu_0$.

2. By Proposition 5.6,

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}.$$

3. For each pair of hypotheses, the rejection region is constructed in the following manner (with cv 's standing for critical values):

- (a) $H_0 : \mu = \mu_0$, $H_A : \mu \neq \mu_0$.

$$\begin{aligned} \frac{\alpha}{2} &= P(\bar{X} < cv_1 | \mu = \mu_0) = P(\bar{X} > cv_2 | \mu = \mu_0) \\ &= P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} < \frac{cv_1 - \mu}{\frac{S}{\sqrt{n}}} | \mu = \mu_0\right) = P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} > \frac{cv_2 - \mu}{\frac{S}{\sqrt{n}}} | \mu = \mu_0\right), \text{ constituting the} \\ &\text{rejection region: } (-\infty, \mu_0 - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}) \cup (\mu_0 + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, +\infty). \end{aligned}$$

- (b) $H_0 : \mu \leq \mu_0$, $H_A : \mu > \mu_0$.

$$\begin{aligned} \alpha &= \max_{\mu' \leq \mu_0} P(\bar{X} > cv | \mu = \mu') \\ &= \max_{\mu \leq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} > \frac{cv - \mu}{\frac{S}{\sqrt{n}}}\right), \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\mu \leq \mu_0} (\mu + t_{\alpha, n-1} \frac{S}{\sqrt{n}}, +\infty) = (\mu_0 + t_{\alpha, n-1} \frac{S}{\sqrt{n}}, +\infty). \end{aligned}$$

- (c) $H_0 : \mu \geq \mu_0$, $H_A : \mu < \mu_0$.

$$\begin{aligned} \alpha &= \max_{\mu' \geq \mu_0} P(\bar{X} < cv | \mu = \mu') \\ &= \max_{\mu \geq \mu_0} P\left(\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} < \frac{cv - \mu}{\frac{S}{\sqrt{n}}}\right), \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\mu \geq \mu_0} (-\infty, \mu - t_{\alpha, n-1} \frac{S}{\sqrt{n}}) = (-\infty, \mu_0 - t_{\alpha, n-1} \frac{S}{\sqrt{n}}). \end{aligned}$$

Method 7.2.4. Confidence interval for variance of a normal population $N(\mu, \sigma^2)$.

1. Similarly, we work on the following three pairs of hypotheses:

- (a) $H_0 : \sigma^2 = \sigma_0^2$, $H_A : \sigma^2 \neq \sigma_0^2$,
(b) $H_0 : \sigma^2 \leq \sigma_0^2$, $H_A : \sigma^2 > \sigma_0^2$, and
(c) $H_0 : \sigma^2 \geq \sigma_0^2$, $H_A : \sigma^2 < \sigma_0^2$.

2. By Proposition 5.6,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

3. For each pair of hypotheses, the rejection region is constructed in the following manner (with cv 's standing for critical values):

(a) $H_0 : \sigma^2 = \sigma_0^2$, $H_A : \sigma^2 \neq \sigma_0^2$.

$$\begin{aligned} \alpha &= P(S^2 < cv_1 | \sigma^2 = \sigma_0^2) = P(S^2 > cv_2 | \sigma^2 = \sigma_0^2) \\ &= P\left[\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)cv_1}{\sigma^2} \mid \sigma^2 = \sigma_0^2\right] \\ &= P\left[\frac{(n-1)S^2}{\sigma^2} > \frac{(n-1)cv_2}{\sigma^2} \mid \sigma^2 = \sigma_0^2\right], \text{ constituting the} \\ &\text{rejection region: } (0, \chi_{1-\frac{\alpha}{2}, n-1}^2 \frac{\sigma_0^2}{n-1}) \cup (\chi_{\frac{\alpha}{2}, n-1}^2 \frac{\sigma_0^2}{n-1}, +\infty). \end{aligned}$$

(b) $H_0 : \sigma^2 \leq \sigma_0^2$, $H_A : \sigma^2 > \sigma_0^2$.

$$\begin{aligned} \alpha &= \max_{\sigma'^2 \leq \sigma_0^2} P(S^2 > cv | \sigma^2 = \sigma'^2) \\ &= \max_{\sigma'^2 \leq \sigma_0^2} P\left[\frac{(n-1)S^2}{\sigma^2} > \frac{(n-1)cv}{\sigma^2} \mid \sigma^2 = \sigma'^2\right], \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\sigma'^2 \leq \sigma_0^2} (\chi_{\alpha, n-1}^2 \frac{\sigma'^2}{n-1}, +\infty) = (\chi_{\alpha, n-1}^2 \frac{\sigma_0^2}{n-1}, +\infty). \end{aligned}$$

(c) $H_0 : \sigma^2 \geq \sigma_0^2$, $H_A : \sigma^2 < \sigma_0^2$.

$$\begin{aligned} \alpha &= \max_{\sigma'^2 \geq \sigma_0^2} P(S^2 < cv | \sigma^2 = \sigma'^2) \\ &= \max_{\sigma'^2 \geq \sigma_0^2} P\left[\frac{(n-1)S^2}{\sigma^2} < \frac{(n-1)cv}{\sigma^2} \mid \sigma^2 = \sigma'^2\right], \text{ constituting the} \\ &\text{rejection region: } \bigcap_{\sigma'^2 \geq \sigma_0^2} (0, \chi_{1-\alpha, n-1}^2 \frac{\sigma'^2}{n-1}) = (0, \chi_{1-\alpha, n-1}^2 \frac{\sigma_0^2}{n-1}). \end{aligned}$$

• Finally, let's go back to the relevant $1 - \alpha$ confidence intervals (of σ^2) in each of the situations above.

- (a) $[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}]$:
 $\sigma_0^2 \in [\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}]$ iff $S^2 \in [\chi_{1-\frac{\alpha}{2}, n-1}^2 \frac{\sigma_0^2}{n-1}, \chi_{\frac{\alpha}{2}, n-1}^2 \frac{\sigma_0^2}{n-1}]$.
- (b) $[\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, +\infty)$:
 $\sigma_0^2 \in [\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, +\infty)$ iff $S^2 \in (0, \chi_{\alpha, n-1}^2 \frac{\sigma_0^2}{n-1}]$.
- (c) $(0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2}]$:
 $\sigma_0^2 \in (0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2}]$ iff $S^2 \in [\chi_{1-\alpha, n-1}^2 \frac{\sigma_0^2}{n-1}, +\infty)$.

- In each of the situations above, the critical value of the parameter falls into the confidence interval iff the sample statistic doesn't fall in the rejection region, i.e., we can't reject the corresponding null hypothesis.
- Particularly, for a situation with one-sided H_0 , we select the corresponding one-sided CI in accordance to the inequality's direction relative to the critical value.
 - * For example, in case (b), consider H_0 as $\sigma_0^2 \geq \sigma^2$ rather than $\sigma^2 \leq \sigma_0^2$.

Revisit - Example 5.A: Normal Population with Known Variance

Consider a sports equipment manufacturer whose 2-lb dumbbells are with weights normally distributed with mean μ and variance 0.0016 (unit: lb). A random sample of 2-lb dumbbells $\{X_i\}_{i=1}^{25}$ with sample size 25 is obtained from the manufacturer, where X_i 's are the weights of the dumbbells. Suppose throughout this problem that $\bar{X} = 2.003$.

1. Write down the null and the alternative hypotheses for a researcher who hopes to test that the weight of the dumbbells is exactly 2 lbs. Can we reject the null at significance level $\alpha = 5\%$?
2. Denote p as the p-value of the above test. Write down an equation that involves the critical value of the sampling distribution used in the test and can pin down the value of p .
3. Write down the null and the alternative hypotheses for a researcher who hopes to test that the weight of the dumbbells is at most 2 lbs. Can we reject the null at significance level $\alpha = 5\%$?
4. Write down the null and the alternative hypotheses for a researcher who hopes to test that the weight of the dumbbells is at least 2 lbs. Can we reject the null at significance level $\alpha = 10\%$?

Note: We will need some of the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Revisit - Example 5.B: Large Sample, Unknown Population

Consider an assembly line of a chip manufacturer whose hourly output is with mean μ and variance σ^2 , both of which are unknown. A random sample of hourly outputs $\{X_i\}_{i=1}^{100}$ with sample size 100 is obtained from the assembly line. Suppose throughout this problem that $\bar{X} = 1500$ and sample variance S^2 equals 400.

Write down the null and the alternative hypotheses for a researcher who hopes to test that the hourly output of the assembly line is at least 1550. Can we reject the null at significance level $\alpha = 10\%$?

Note: We will need some of the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.

Revisit - Example 5.C: Small Sample, Normal Population with Unknown Variance

Consider a pizza restaurant whose 10-inch pizzas are with diameters normally distributed with mean μ and variance σ^2 (unit: inch), both of which are unknown. A random sample of 10-inch pizzas $\{X_i\}_{i=1}^{16}$ with sample size 16 is obtained from the restaurant, where X_i 's are the diameters of the pizzas. The sample mean $\bar{X}$ equals 10.2 and sample variance S^2 equals 0.04.

1. Write down the null and the alternative hypotheses for a researcher who hopes to test that the diameter of the pizzas is no higher than 9.8 inches. Can we reject the null at significance level $\alpha = 5\%$?

Note: We will need some of the following t critical values in this problem: $t_{0.1,15} = 1.3406$, $t_{0.05,15} = 1.7531$, $t_{0.025,15} = 2.1314$.

2. Write down the null and the alternative hypotheses for a researcher who hopes to test that the variance of the diameter of the pizzas is exactly 0.1 (unit: inch). Can we reject the null at significance level $\alpha = 5\%$?

Note: We will need some of the following χ^2 critical values in this problem: $\chi_{0.05,15}^2 = 24.9958$, $\chi_{0.1,15}^2 = 22.3071$, $\chi_{0.9,15}^2 = 8.5468$, $\chi_{0.95,15}^2 = 7.2609$.

3. Is our conclusion in part 2 consistent with that from the confidence interval method?

Revisit - Example 6.B: Large sample, Bernoulli population

Consider a university where the students are either for or against a certain university-wide regulation. The fraction of students who are for the regulation is p . A random sample of students $\{X_i\}_{i=1}^{400}$ with sample size 400 is obtained from the students in the university, where for each student with label $i \in \{1, 2, \dots, 400\}$, $X_i = 1$ if the student is for the regulation and $x_i = 0$ if the student is against the regulation. The sample mean $\bar{X}$ equals 0.8.

Write down the null and the alternative hypotheses for a researcher who hopes to test that the fraction of students that are for the regulation is exactly 0.9. Can we reject the null at significance level $\alpha = 5\%$?

Note: We will need some of the following z critical values in this problem: $z_{0.1} = 1.2816$, $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$.